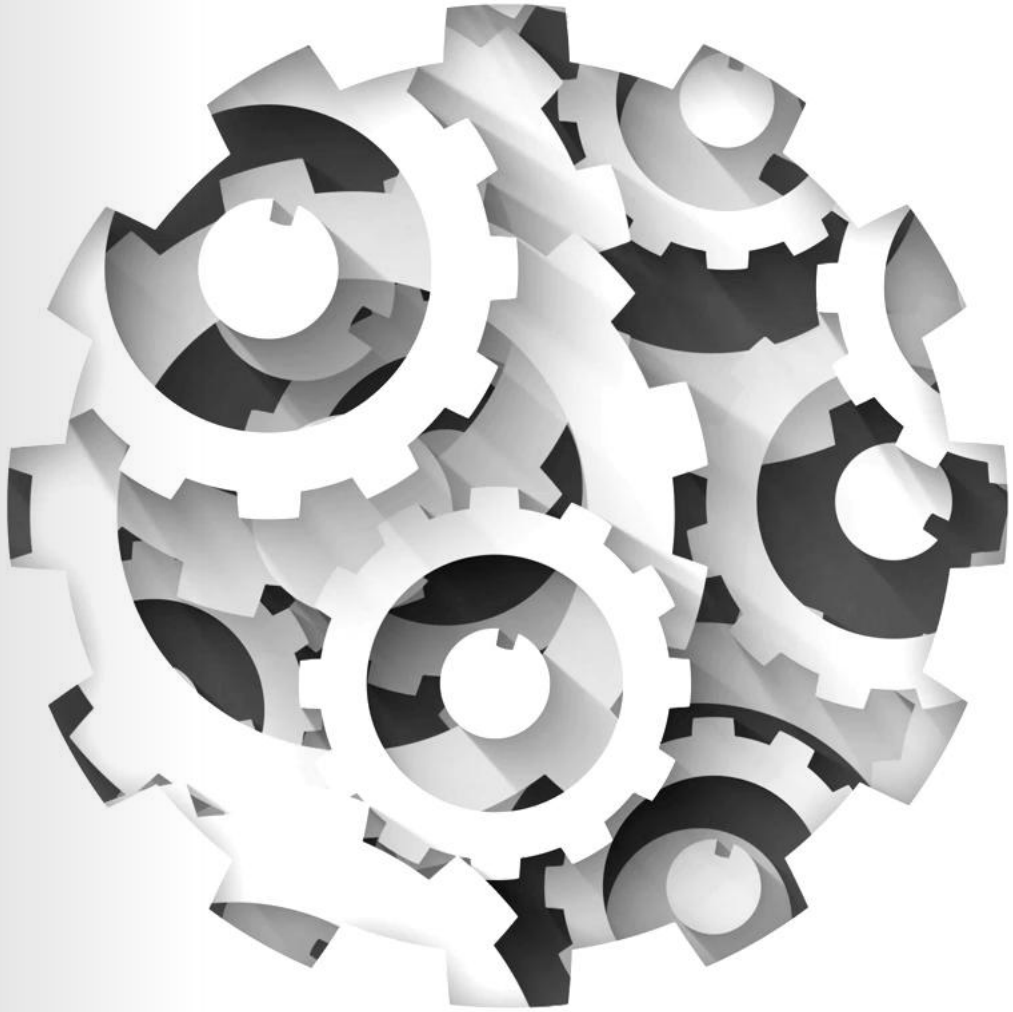
Project 6

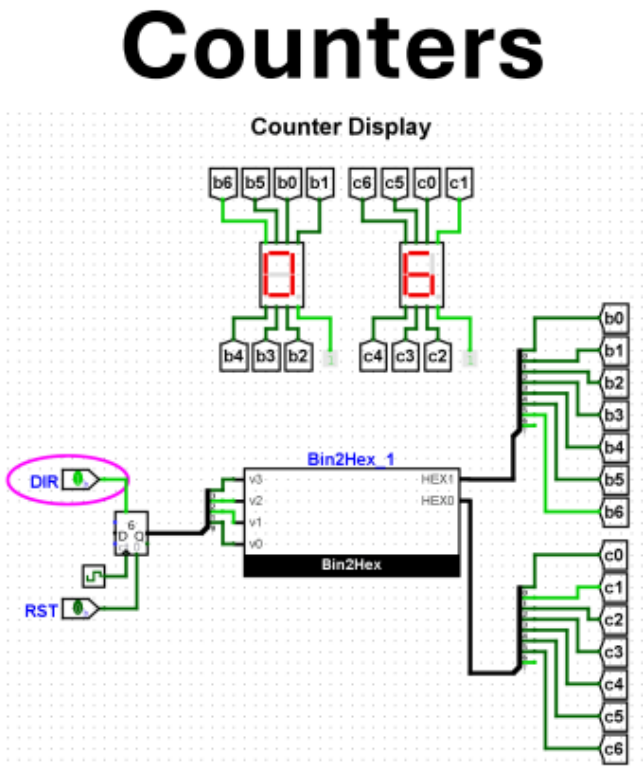


By: Nicklaus Brown

What's given

Building a 10 to 0 clock counter going down based of this main circuit given in lab.

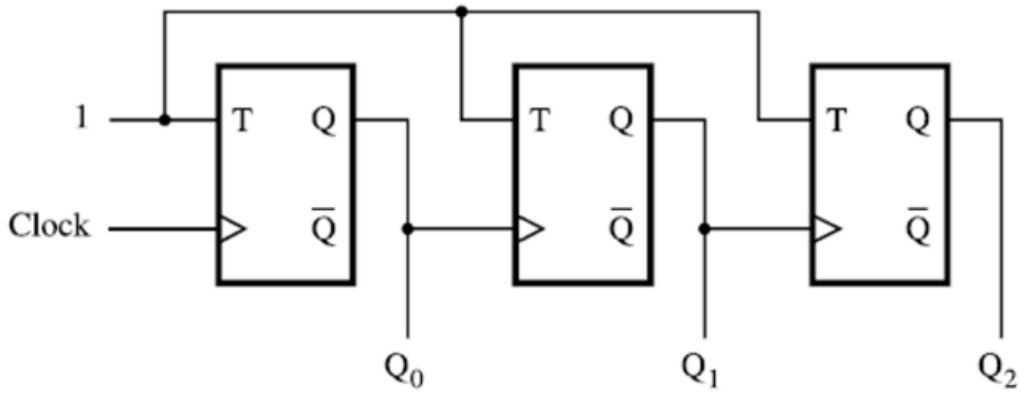
We are given multiple templates that can be used to create this function but most importantly using t flip flops.



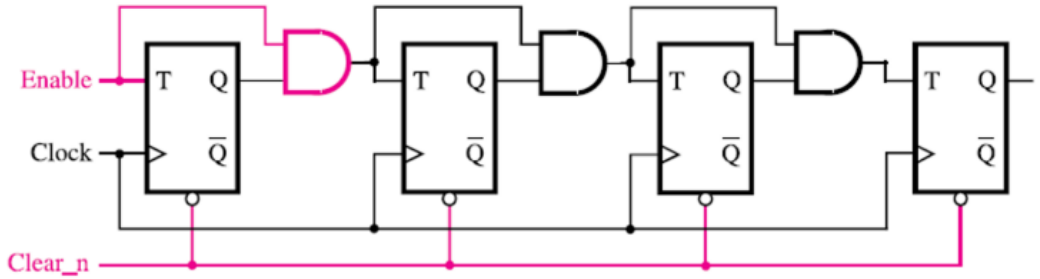
Templates I thought would be used

- Two main templates that were provided during that lab that I took inspiration from was the down counter template and the counter with enable and clear template for there good important functions

Down Counter Template



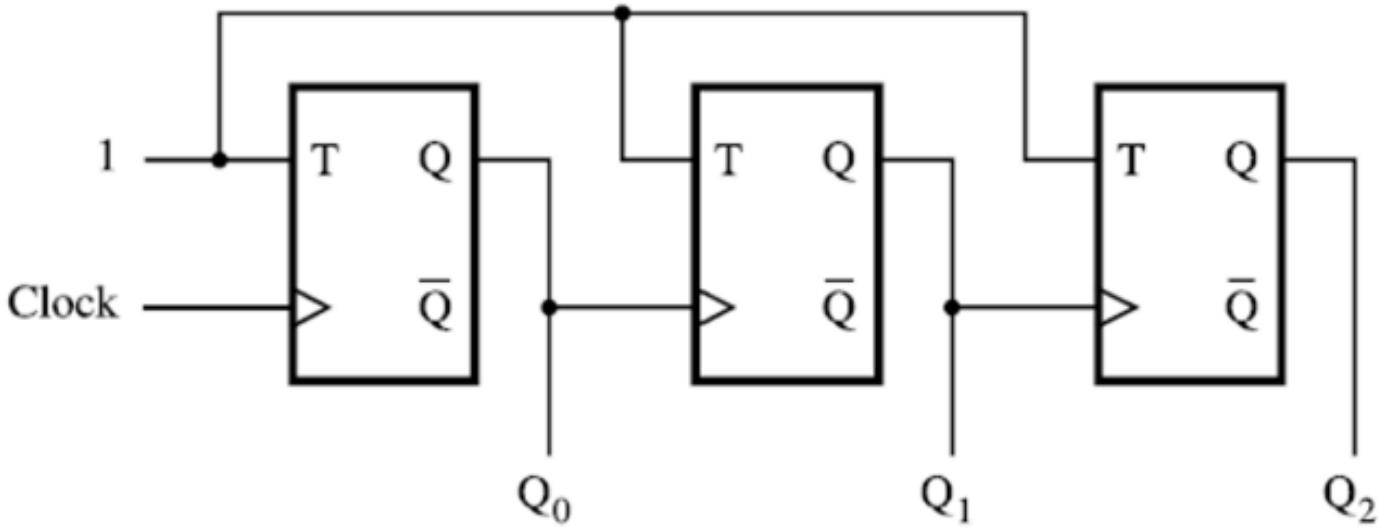
Counter with Enable and Clear Template



Reasons

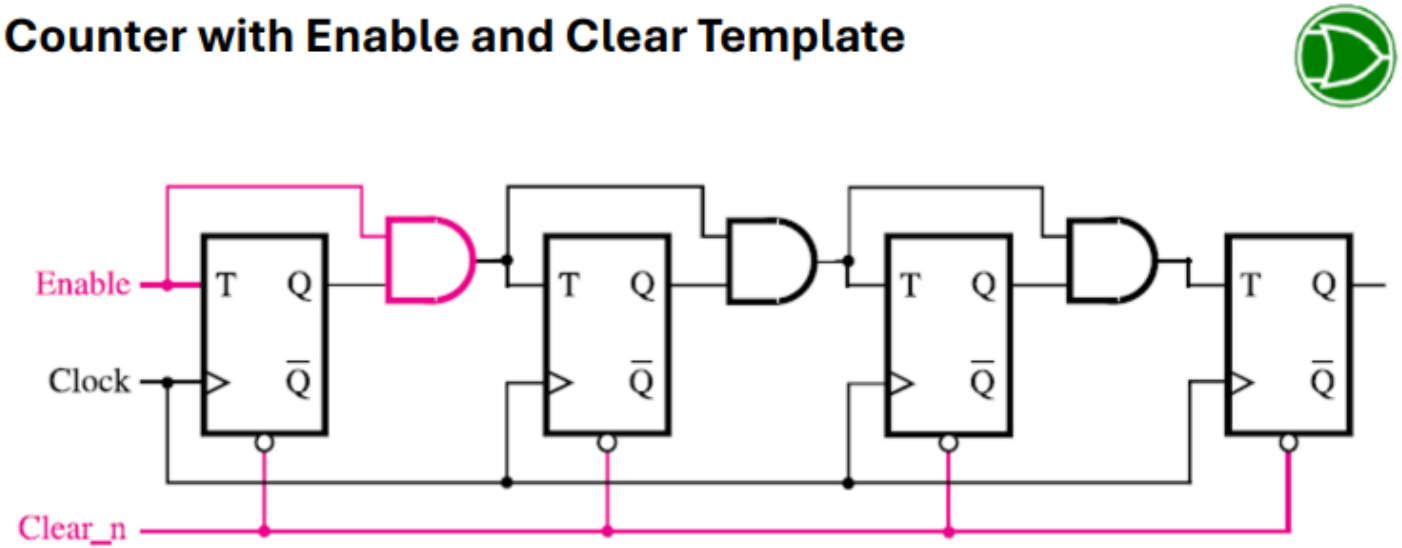
The Down Counter template is self explanatory as it provides the basic function of counting down from 15 to 0 and resetting. This down counter is really important as you need the inputs to work effectively with other templates that utilize t flip flops in different ways. So the next modification for the next step in my circuit building was resting the counter to 10 again.

Down Counter Template



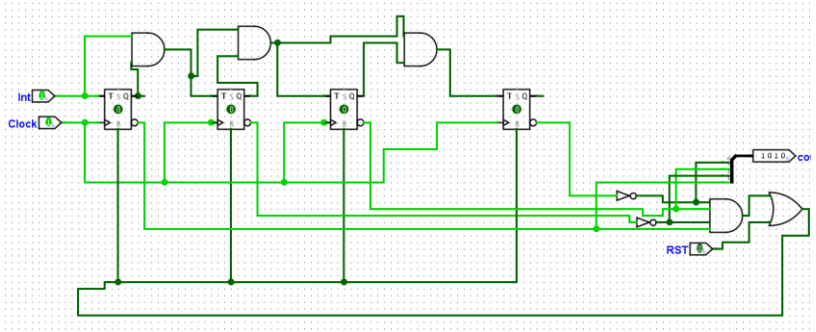
Enable and clear template

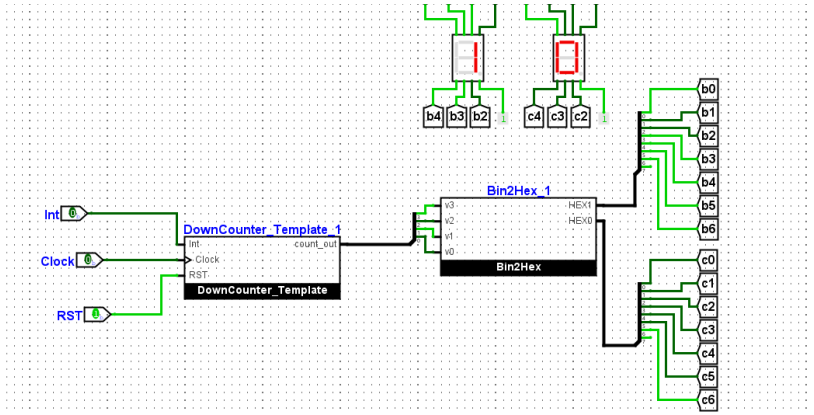
- The next template I took inspiration from was the enable and clear template as it was based off AND gate security that made sure the starting sequence would start from 10. when the counter reached 0. Which was a success for my circuit.



My final circuit design

- The countdown circuit I designed during lab focuses a lot on resetting to 10 when the count reaches 0. However there has been flaws with my design and the goal I wanted to achieve as I have neglected a huge import piece while designing it

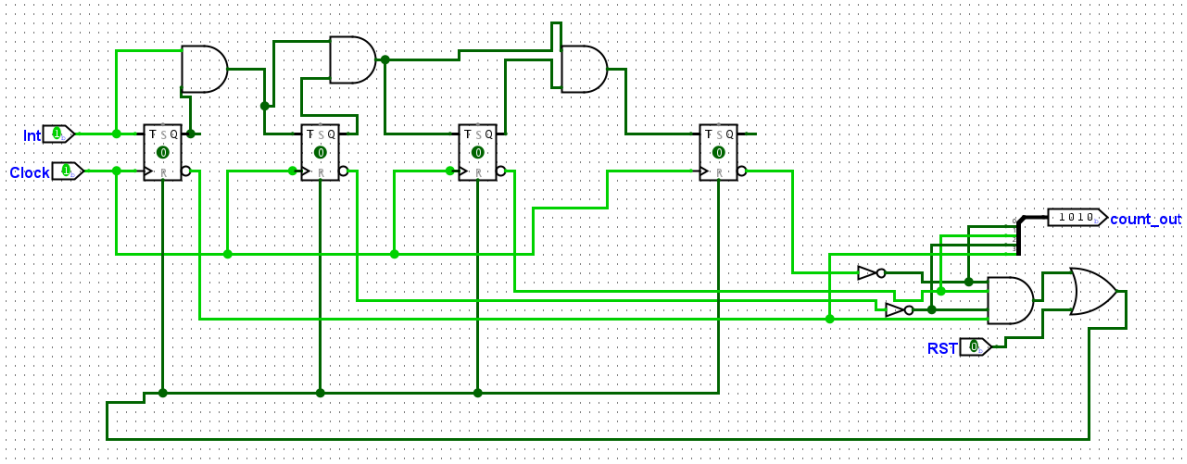




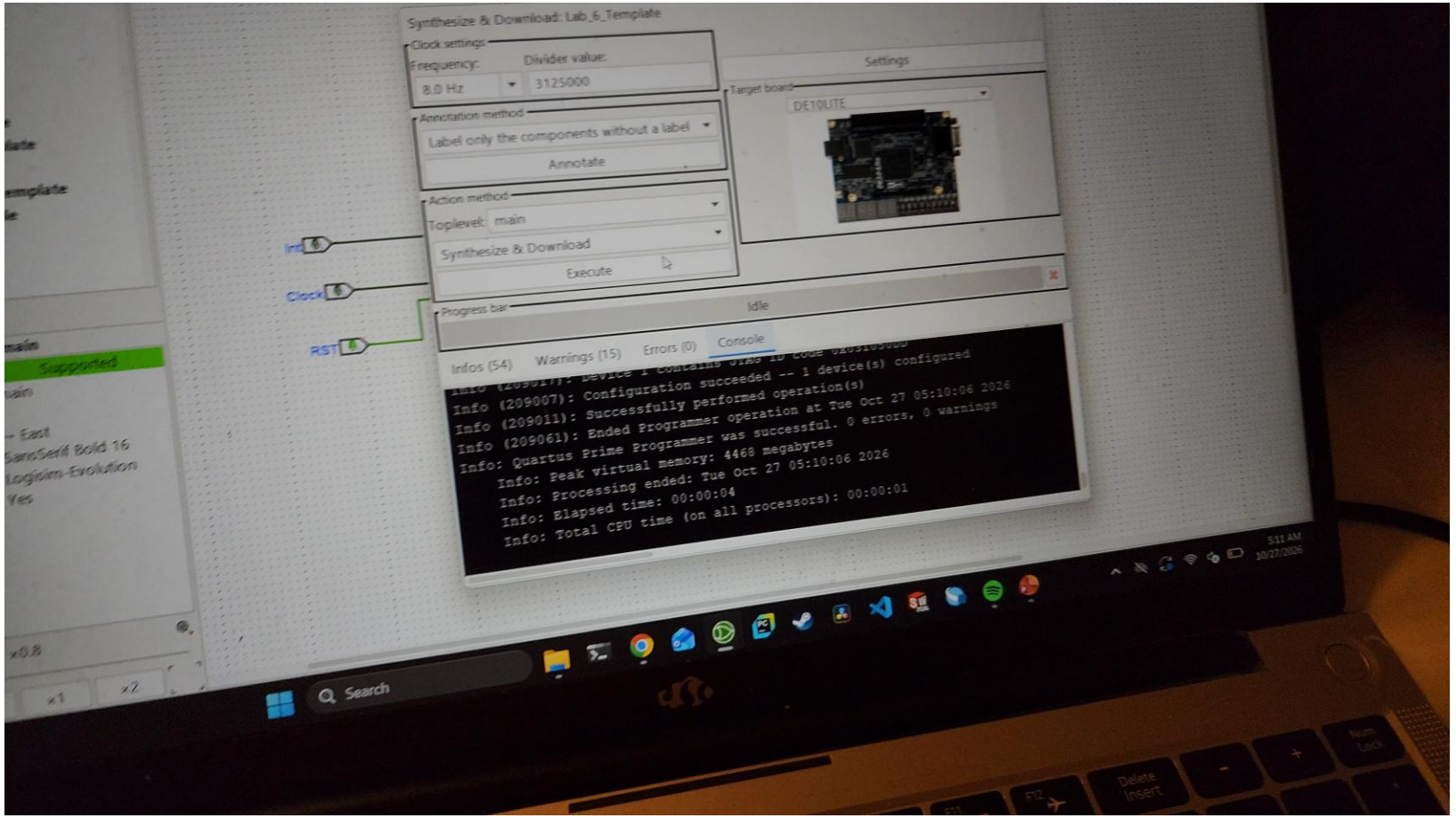
Errors and fixes

From my circuit testing, its clear that there's no input to force the counter to go down. I am having trouble with implementing other templates into the design to make the counter go down while also making sure the reset counter would work.

Because of this error, it seems that the reset back to 10 is so effective that any chance it does go down, the number instantly goes back to 10.



Video of the outcome



Fixes for the future

- After finishing and attempting this lab, I can say that the next most important step would be looking more into the flip flops that the lab is centered around. As it's an important the the inputs are driven by a inverted Q and ensuring that the clock signal and t flip flops share the same signal so it doesn't automatically reset as seen in my video. Creating better more effective inputs for both the down counter and clear templates would result in a more effective clock counter.